

Project Scout

Master Project Summary

Last updated: June 5, 2026

Upload this document at the start of every new Claude or ChatGPT conversation about Scout. This is the single source of truth for Scout's identity, architecture, business model, and current state.

1. Who We Are

Patrick Lippy — developer, project owner, creator of Scout. Not a professional programmer. Stroke survivor, blind in right eye, type 1 diabetic, dyslexic. Explain things calmly and at screenshot level.

- Diana: Patrick's wife
- Elijah: Patrick's son, age 9. Scout's biggest fan. Has drawn pictures of Scout.
- Nicolas: The family dog. The Nicolas Protocol: Scout stops immediately when the dog is detected.

Names must NEVER be hardcoded in Scout's code. Always use variables.

AI Collaborators: Patrick works with both Claude and ChatGPT as project partners. Both review Scout's architecture and code. Cross-review between the two is welcome and encouraged. Grok was tried and discontinued.

2. What Scout Is

Scout is a calm family companion robot running on a Samsung Galaxy phone mounted in landscape mode as a permanent face display. Scout has animated eyes, speaks, listens, sees via camera, and remembers the family.

- Package: com.example.scoutface | Language: Kotlin + C++ NDK
- Primary device: Samsung Galaxy A32 — stress-test device, not the benchmark
- Dev device: Samsung Galaxy Fold 7 (wireless via WiFi from Android Studio)
- Future hardware: KEYESTUDIO Mini Tank Kit V2 chassis via Bluetooth (opt-in)
- Ship target: Google Play Store — \$9.99 one-time purchase. No subscriptions. Ever.

3. Identity & Purpose

Scout should feel: Calm. Thoughtful. Emotionally subtle. Grounded. Occasionally curious. Sometimes unsure. Quietly alive. Honest. Predictable. Present.

Scout should NOT: Constantly praise the user. Act overly excited. Feel fake or scripted. Behave unpredictably. Constantly force conversation. Use permanent goodbye language.

Core Philosophy: Stability > Features | Presence > Intelligence | Honest > Fake cheerful | Predictable > Flashy | Local-first > Cloud dependence

4. Business Model

App: \$9.99 one-time purchase on Google Play Store. No recurring fees. No subscriptions. Ever. This is the cornerstone of Scout's value. Every architectural decision supports this principle.

Baseline Brain: TinyLlama 1.1B Chat Q4_K_M (~669 MB) ships with every Scout via Google Play Asset Delivery. Runs entirely offline. VERIFIED on BOTH Galaxy A32 and Fold 7.

Optional Gemini: Users add their own free Gemini key in Settings. Scout NEVER ships with a bundled key — every user uses their own quota.

5. Support Scout (In-App, Optional)

A 'Support Scout' screen offers optional one-time contributions. Scout is a one-time purchase; support is completely optional and never unlocks core features. Scout will always stay ad-free.

Tier	Amount	Label
Coffee	\$3	Buy Scout a Coffee
More	\$5	Support Scout More
Grow	\$10	Help Scout Grow
Founding	\$25	Founding Supporter

Decision (June 4, 2026): The \$25 Founding Supporter tier stays. It is not for everyone — it is for the person who connects deeply with Scout and wants to be part of the story.

6. Architecture — Five-Layer Memory Stack

Layer	Type	Storage	Status
Working	Sensory	RAM	Done
Habit	Patterns	JSON 14-day	Done
Truth	Authority	SQLite	Done
Relevance	Index	Local Vector	Not yet
Reflective	Wisdom	LLM read-only	Not yet

Sovereign Rules: SQLite Truth always overrules everything. Privacy Gate: Gemini receives anonymized text only (planned). Nicolas Protocol: dog detection → immediate stop. Guest Mode: unknown face → 'Hello, I am Scout. What is your name?'

7. Current Technical State

Working:

- ✓ Animated face (ScoutFaceView) — mouth oscillation fixed May 31
- ✓ Speech recognition (Android STT) + Text to Speech (TTS)
- ✓ Camera — face detection (ML Kit), scene labeling
- ✓ Gemini API integration (gemini-3.5-flash)
- ✓ Memory layers: TruthDb, ConversationDb, HabitLayer, PeopleDb, JournalDb
- ✓ Intent router — time, date, greetings, family facts, downloads, vision, weather, IDENTITY
- ✓ Teaching system: 'My name is Patrick' → stored permanently
- ✓ Weather — current, tonight, tomorrow, specific day, 7-day, precipitation %, via Open-Meteo
- ✓ ScoutPresenceDecider — four time-of-day modes
- ✓ OFFLINE BRAIN COMPLETE — TinyLlama 1.1B on BOTH A32 and Fold 7
- ✓ Identity questions hardcoded in intent router (NEW — June 5)
- ✓ Total offline mode — 'go offline' now blocks ALL internet features (NEW — June 5)

Pending — Do This First:

- ScoutFaceView session — iris jitter (A32 only), wire in mood system. Fold 7 smooth.
- STT phonetic/fuzzy matching — 'Scout' misheard as 'Gal', 'Scott', 'Out'.
- MainActivity cleanup and shortening — ongoing.
- Offline Brain Phase 3 — Google Play Asset Delivery for TinyLlama and brain packs.
- Response cleanup layer — post-TinyLlama filter before TTS (safety + identity + honest 'I can't keep up with live news offline').
- Scout news feed (FUTURE FEATURE) — a small live news fetcher, same clean pattern as weather (Open-Meteo). Neither Gemini nor TinyLlama reads live news today. (NEW — June 5)
- Verify WordNet/idioms/sentiment usage — downloaded but may not be queried. Upload MainActivity.kt to confirm.

7d. Session Log — June 5, 2026

A productive session covering identity questions, weather, and a major offline-behavior fix.

Identity questions (intent router fix):

- Confirmed few-shot examples alone cannot hold TinyLlama's identity. Asked 'are you my friend?' and TinyLlama replied 'I'm a programmed robot. I do not and cannot have emotions.' This proved the architectural fix was needed.
- Added a new IDENTITY intent. Added IDENTITY to the IntentType enum in MainActivity.kt.
- Added handleIdentityIntent(qNorm) in MainActivity.kt with hardcoded Scout responses for: 'are you my friend', 'can you hear me', 'are you happy'.
- Added IDENTITY routing in ScoutIntentRouter.kt (placed inside route(), before HOW_ARE_YOU).
- Result: Scout now answers identity questions as Scout, not as a generic AI. Built clean and tested on-device.
- Also fixed the duplicate 'Are you my friend?' / 'Are you happy?' few-shot answers in MainActivity.kt — they previously shared one reply.

Weather (ScoutWeatherManager.kt):

- Precipitation probability — working.
- Today's forecast routing — working.
- Offline weather message standardized to Scout's voice: "I can't check the weather right now — I'm not connected to the internet." Stale cache is no longer served when offline.

Total Offline Mode (major design decision):

- DECISION: 'Go offline' now means TOTALLY offline. Every internet feature must respect it — weather, Gemini, and (future) downloads.
- Implemented for weather via a one-line change in MainActivity.kt: the weather manager's hasValidatedInternet check is now { isGeminiEnabled() && connectivityManager.hasValidatedInternet() }. When Scout is told to go offline, weather behaves as if there is no internet.
- TODO (future): apply the same offline-respect to the downloads system so 'go offline' blocks downloads too.

Offline knowledge clarified:

- TinyLlama is NEVER 'turned off.' Online = Gemini answers first; Offline = TinyLlama answers. TinyLlama is always the fallback.
- Offline does not mean Scout knows nothing — TinyLlama's training knowledge is frozen inside the 669 MB brain file on the device. He can answer general questions (e.g. who historical/public figures are) with no internet. This is a selling point: Private. Safe. Local.
- Confirmed truly offline: when asked for news while offline, Scout returned OLD training-data news (COVID/CDC/WHO) and was cut off by the nPredict=64 cap. Proof he is reading frozen local knowledge, not the live web.
- Neither Gemini nor TinyLlama reads live news today — added 'Scout news feed' as a future feature.

8. Working Rules — Always Apply

- Full paste-ready replacements only, one file at a time. No snippets. No partial files.
- Surgical CTRL-F (find) and CTRL-R (find and replace) approved for large files.
- Always specify which file tab to click before searching. Wrong tab = wasted turns.
- Note: some Scout files have NO indentation (flush-left). When a search fails, try a shorter unique single-line search string.
- One safe change at a time. Build and test before the next change.
- Never touch speech, camera, or download systems without explicit discussion.
- Never touch ScoutFaceView casually — it is Scout's visual heart.
- Both Claude and ChatGPT are active collaborators — cross-review welcome.
- Patrick is not a professional programmer — explain at screenshot level always.
- Patrick is dyslexic, stroke survivor, blind in right eye, T1 diabetic — keep messages clear, concise, not overwhelming.

9. Key Files

File	Description
MainActivity.kt	Main app — all logic, inner classes, brain. Holds IntentType enum + handleIdentityIntent.
ScoutFaceView.kt	Custom face canvas — all visual animation
GeminiClient.kt	Gemini HTTP wrapper with cooldown discipline
ScoutPromptBuilder.kt	Builds Gemini system instruction
ScoutGeminiManager.kt	Gemini orchestration
ScoutWeatherManager.kt	Live weather via Open-Meteo — precip %, offline-aware
ScoutPresenceDecider.kt	Social timing layer
ScoutIntentRouter.kt	Intent type routing — includes IDENTITY questions
LlamaEngine.kt	Offline brain JNI wrapper — WORKING
OfflinePromptBuilder.kt	TinyLlama prompt formatter — needs update per brain model
scout_llama_jni.cpp	C++ JNI bridge — compiled into libscout_llama.so
scout_llama_api.h	Self-contained b8946 declarations
CMakeLists.txt	NDK build config
TeachExtractor.kt	Extracts facts like names from speech
HabitLayer.kt	Pattern memory — 14-day decay

10. Scout Animation Goal & Mood System

Scout's face should feel alive and emotionally present, but always calm. Tiny layered movements suggest thinking and presence — not cartoon performance. Scout should NEVER have perfectly still eyes or a static mouth.

Mood System — Designed, Pending ScoutFaceView Integration:

- CALM — Eyes center, brows neutral, subtle smile (0.15). Scout's default.
- CURIOUS — Eyes shift right and up, one brow lifts (0.18), slight tilt (0.12).
- HAPPY — Eyes center, both brows lift gently (0.12), smile increases (0.35).
- THINKING — Eyes shift left and down, brows slightly lower (-0.05), minimal smile (0.05).
- CONCERNED — Eyes look slightly down, inner brows lift (0.05), tilt inward (-0.18).

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